

Practice Set 9

MA628 - Financial Derivatives Pricing

April 20, 2026

1. Compute the stochastic differentials dY when

- (a) $Y(t) = \exp(\beta t)$
- (b) $Y(t) = \exp(\beta B(t))$
- (c) $Y(t) = \int_0^\infty g(s)dB(s)$, where g is adapted process.
- (d) $Y(t) = \exp(\beta X(t))$ and $dX(t) = \mu dt + \sigma dB(t)$.
- (e) $Y(t) = 1/X(t)$ where $dX(t) = \alpha X(t)dt + \sigma X(t)dB(t)$.

2. Compute the stochastic differential dY for

- (a) $Y(t) = \int_0^t e^{-as} \sigma dB(s)$.
- (b) $Y(t) = e^{at} \int_0^t e^{-as} \sigma dB(s)$.

3. Let X be the solution of the SDE

$$dX(t) = \alpha X(t)dt + \sigma dB(t), \quad X(0) = x_0.$$

- (a) Find $X(t)$
- (b) Find $\mathbb{E}[X(t)]$
- (c) Find $\text{Var}(X(t))$.

4. Let X and Y satisfy the following systems of SDEs

$$dX(t) = \alpha X(t)dt + Y(t)dB(t), \quad X(0) = x_0; \quad dY(t) = \alpha Y(t)dt - X(t)dB(t), \quad Y(0) = y_0.$$

- (a) Show that R , defined by $R(t) = X^2(t) + Y^2(t)$ is deterministic.
- (b) Compute $\mathbb{E}X(t)$ and $\text{Cov}(X(t), Y(t))$.

5. Let X and Y satisfy the following systems of SDEs

$$dX(t) = \alpha X(t)dt + \beta X(t)dB_1(t), \quad X(0) = x_0; \quad dY(t) = \gamma Y(t)dt + \delta X(t)dB_2(t), \quad Y(0) = y_0,$$

where α, β, γ and δ are constants and B_1 and B_2 are independent Brownian motions. Compute $\mathbb{E}[X(t)Y(t)]$.